

Large vector-valued fibers over arbitrary target balls

A candidate partial answer for arXiv:1902.01387

Autonomous proof attempt, lane 6

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Status: candidate partial result, likely valid. The result answers the source question for $X = c_0$ and arbitrary Banach target space Y , and for a corresponding family of fibers when X contains a 1-complemented isometric copy of c_0 . It does not settle the question for a general infinite-dimensional source space X .

1 Source question

For Banach spaces X, Y , let $\mathcal{M}_\infty(B_X, B_Y)$ be the nonzero algebra homomorphisms from $\mathcal{H}^\infty(B_X)$ to $\mathcal{H}^\infty(B_Y)$. The canonical projection

$$\xi : \mathcal{M}_\infty(B_X, B_Y) \longrightarrow \overline{B}_{\mathcal{H}^\infty(B_Y, X^{**})}$$

gives the fiber $\mathcal{F}(g) = \xi^{-1}(g)$. Dimant and Singer ask whether, for infinite-dimensional X , the fibers over nonconstant maps

$$g \in \overline{B}_{\mathcal{H}^\infty(B_Y, X^{**})}, \quad g(B_Y) \subset B_{X^{**}},$$

are large. They emphasize the unresolved norm-one case: can their analytic disk theorem be extended to $\|g\| = 1$ while every value $g(y)$ remains in the open ball? See page 15 of the source PDF and Figure 1 [1].

A later paper proves that every fiber of $\mathcal{M}_\infty(B_{c_0}, B_{c_0})$ contains 2^c disjoint analytic Gleason-isometric copies of $B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$ [2, Theorem 2.4]. The target domain in that theorem is again c_0 . The first result below removes that restriction completely.

2 Proof intuition

The proof of the later c_0 theorem needs a common subsequence of the coordinate functions g_n . Its published selection lemma invokes a compactness theorem for bounded holomorphic functions on B_{c_0} , which is the only step apparently tied to the target domain $Y = c_0$.

The replacement is one-dimensional. Select a subsequence for which the single scalar sequence $g_n(0)$ converges. If its limit lies inside the disk, Schwarz–Pick bounds every selected coordinate uniformly below one at each fixed $y \in B_Y$. If the limit lies on the circle, the same inequality forces pointwise convergence to that unimodular limit throughout B_Y . This is exactly the dichotomy needed by the original construction, for every Banach space Y . No compactness or separability of Y is required.

Finally, if c_0 is 1-complemented in a source space X , restriction from B_X to that copy of B_{c_0} is a norm-one quotient on the holomorphic algebras and has a norm-one right inverse. Precomposing homomorphisms with this restriction preserves the fiber, analyticity, and the Gleason metric.

3 The coordinate dichotomy

Write

$$\rho(a, b) = \left| \frac{a - b}{1 - \bar{b}a} \right| \quad (a, b \in \mathbb{D})$$

for the pseudo-hyperbolic metric.

Lemma 1 (target-domain-free coordinate dichotomy). *Let Y be any complex Banach space and let $(u_n) \subset \overline{B}_{\mathcal{H}^\infty(B_Y)}$. There is a subsequence (u_{n_k}) for which one of the following holds:*

- (i) *for every $y \in B_Y$, $\sup_k |u_{n_k}(y)| < 1$;*
- (ii) *there is $\lambda \in \mathbb{T}$ such that $u_{n_k}(y) \rightarrow \lambda$ for every $y \in B_Y$.*

Proof. Pass to a subsequence, still denoted (u_n) , such that $u_n(0) \rightarrow \lambda \in \overline{\mathbb{D}}$.

Suppose first that $|\lambda| < 1$. Choose q with $|\lambda| < q < 1$ and discard finitely many terms so that $|u_n(0)| \leq q$. Schwarz–Pick for the holomorphic map $u_n : B_Y \rightarrow \mathbb{D}$ gives

$$\rho(u_n(y), u_n(0)) \leq \|y\|.$$

Consequently, with $r = \|y\| < 1$,

$$|u_n(y)| \leq \frac{q + r}{1 + qr} < 1$$

for every selected n . This proves (i).

Suppose now that $|\lambda| = 1$. For fixed $y \in B_Y$, the same Schwarz–Pick inequality keeps $u_n(y)$ within pseudo-hyperbolic distance $\|y\| < 1$ of $u_n(0)$. A sequence in $\mathbb{D}$ at a fixed pseudo-hyperbolic distance strictly below one from a sequence converging to $\lambda \in \mathbb{T}$ must itself converge to λ : after passing to a cluster point $a \in \overline{\mathbb{D}}$, the limiting inequality would be $|a - \lambda| \leq r|1 - \bar{\lambda}a|$; the two absolute values are equal when $|\lambda| = 1$, so $r < 1$ forces $a = \lambda$. Thus $u_n(y) \rightarrow \lambda$ for every y , proving (ii). $\square$

4 Main partial theorem

We use the notation and Gleason metric of [1, 2]. In particular, $\mathcal{F}_Y(g)$ denotes the fiber in $\mathcal{M}_\infty(B_{c_0}, B_Y)$.

Theorem 2 (arbitrary target domain). *Let Y be any complex Banach space. For every*

$$g \in \overline{B}_{\mathcal{H}^\infty(B_Y, \ell_\infty)}$$

there is a map

$$\Psi_g : (\beta\mathbb{N} \setminus \mathbb{N}) \times B_{\mathcal{H}^\infty(B_Y, \ell_\infty)} \longrightarrow \mathcal{F}_Y(g)$$

such that:

- (1) *for every $\eta \in \beta\mathbb{N} \setminus \mathbb{N}$, the map $\Psi_g^\eta(h) = \Psi_g(\eta, h)$ is an analytic Gleason isometry;*
- (2) *if $\eta_1 \neq \eta_2$, the images of $\Psi_g^{\eta_1}$ and $\Psi_g^{\eta_2}$ lie in different Gleason parts.*

In particular, every fiber contains 2^c pairwise disjoint analytic copies of the full ball $B_{\mathcal{H}^\infty(B_Y, \ell_\infty)}$.

Proof. Write $g = (g_n)_n$, with $g_n \in \overline{B_{\mathcal{H}^\infty(B_Y)}}$. Apply Lemma 1. This supplies precisely the two alternatives of [2, Lemma 2.2], now on B_Y : either some coordinate subsequence $g' = (g_{n_k})_k$ satisfies $g'(B_Y) \subset B_{\ell_\infty}$, or a coordinate subsequence converges pointwise to a number $\lambda \in \mathbb{T}$.

We spell out why the remainder of the proof of [2, Theorem 2.4] is independent of the target domain. Let $\mathbb{J} = \{n_k : k \in \mathbb{N}\}$. The scalar construction in [2, Theorem 1.1 and its proof] provides maps $\Phi_{z, \mathbb{J}}^k(w)$ in the scalar spectrum of $\mathcal{H}^\infty(B_{c_0})$. In the first alternative define, for $h \in B_{\mathcal{H}^\infty(B_Y, \ell_\infty)}$,

$$\Psi_g^k(h)(f)(y) = \Phi_{g(y), \mathbb{J}}^k(h(y))(f), \quad f \in \mathcal{H}^\infty(B_{c_0}), \quad y \in B_Y.$$

For fixed k, h, f , the two-variable map obtained by putting $g(y)$ and $h(y')$ in the scalar formula is separately holomorphic on $B_Y \times B_Y$; Hartogs' theorem and restriction to the diagonal show that $\Psi_g^k(h)(f)$ is holomorphic. The scalar construction bounds it by $\|f\|$, so it belongs to $\mathcal{H}^\infty(B_Y)$.

The general weak-star compactness of $\mathcal{M}_\infty(B_{c_0}, B_Y)$, established via the canonical predual $\mathcal{G}^\infty(B_Y)$ in [1, Section 2], lets $k \mapsto \Psi_g^k(h)$ extend to $\beta\mathbb{N}$. For $\eta \in \beta\mathbb{N} \setminus \mathbb{N}$ the scalar evaluation identity is

$$\delta_y \circ \Psi_g^\eta(h) = \Phi_{g(y), \mathbb{J}}^\eta(h(y)). \quad (1)$$

It gives $\xi(\Psi_g^\eta(h)) = g$. The scalar maps are Gleason isometries, so taking the supremum of their pseudo-hyperbolic equalities over $y \in B_Y$ proves that Ψ_g^η is a Gleason isometry. Distinct ultrafilters give scalar images in different Gleason parts; (1) transfers this separation to the vector-valued spectrum. Analyticity in h follows pointwise from the analytic scalar maps and then from the general point-evaluation criterion for locally bounded maps into $\mathcal{H}^\infty(B_Y)$ [1, Lemma 2.3].

In the second dichotomy alternative, if $\lambda \neq 1$, then for every fixed y the selected coordinates $g_{n_k}(y)$ are eventually a positive distance from 1. This is exactly the second case of the proof of [2, Theorem 2.4], and the preceding construction applies. If $\lambda = 1$, multiply the selected coordinates by a fixed unimodular scalar different from 1 and use the coordinate-rotation Gleason isometry of [2, Lemma 2.3]. These operations concern only the c_0 source coordinates and are unchanged when B_{c_0} in the target variable is replaced by B_Y . This proves both assertions. $\square$

Corollary 3 (the c_0 case of the source question). *For every Banach space Y , every nonconstant $g \in \overline{B_{\mathcal{H}^\infty(B_Y, \ell_\infty)}}$ satisfying $g(B_Y) \subset B_{\ell_\infty}$ has a large fiber in the strongest sense of Theorem 2. This includes all such maps with $\|g\| = 1$.*

Proof. Apply Theorem 2; its conclusion holds for every g in the closed ball, without requiring either nonconstancy or pointwise interiority. $\square$

5 Transfer through a complemented copy of c_0

Theorem 4 (complemented- c_0 transfer). *Let X, Y be complex Banach spaces. Suppose there are norm-one maps*

$$J : c_0 \longrightarrow X, \quad P : X \longrightarrow c_0, \quad PJ = I_{c_0},$$

*with J an isometry. For every $g_0 \in \overline{B_{\mathcal{H}^\infty(B_Y, \ell_\infty)}}$, the fiber in $\mathcal{M}_\infty(B_X, B_Y)$ over $J^{**} \circ g_0$ contains the images furnished by Theorem 2, with analyticity and all Gleason distances preserved. Hence it contains 2^c disjoint analytic Gleason-isometric copies of $B_{\mathcal{H}^\infty(B_Y, \ell_\infty)}$.*

Proof. Define

$$R : \mathcal{H}^\infty(B_X) \rightarrow \mathcal{H}^\infty(B_{c_0}), \quad Rf = f \circ J, \quad E : \mathcal{H}^\infty(B_{c_0}) \rightarrow \mathcal{H}^\infty(B_X), \quad Eu = u \circ P.$$

Both are norm-one unital algebra homomorphisms, $RE = I$, and E is an isometry. For $\Theta \in \mathcal{M}_\infty(B_{c_0}, B_Y)$ put

$$L\Theta = \Theta \circ R \in \mathcal{M}_\infty(B_X, B_Y).$$

If $\Theta \in \mathcal{F}_Y(g_0)$ and $x^* \in X^*$, then

$$(L\Theta)(x^*)(y) = \Theta(J^*x^*)(y) = g_0(y)(J^*x^*) = (J^{**}g_0(y))(x^*).$$

Thus $L\Theta$ lies in the fiber over $J^{**} \circ g_0$.

The equality $RE = I$ makes L injective. More precisely, for scalar characters φ, ψ on $\mathcal{H}^\infty(B_{c_0})$, the unit ball under R is exactly the unit ball of $\mathcal{H}^\infty(B_{c_0})$: one inclusion follows from $\|R\| = 1$, and the reverse inclusion follows by applying E . Therefore

$$\rho(\varphi \circ R, \psi \circ R) = \rho(\varphi, \psi).$$

Taking the supremum over $y \in B_Y$ shows that L preserves the vector-valued Gleason metric. Since L is bounded linear on the ambient operator spaces, it also preserves analyticity. Apply L to all the copies from Theorem 2. $\square$

6 Verification, scope, and novelty status

The new mathematical input is Lemma 1; every inequality in it is scalar Schwarz–Pick. Theorem 2 then audits the published proof step by step: scalar c_0 -fiber maps, Hartogs, weak-star compactness, the evaluation identity, and the point-evaluation holomorphy criterion are all stated for general Banach target domains or depend only on the c_0 source. Theorem 4 is an exact algebraic restriction/right-inverse argument.

This is only a partial answer to arXiv:1902.01387. It does not construct coherent analytic families in scalar fibers for a general infinite-dimensional X . The known fact that each individual scalar fiber is large supplies no such dependence on $g(y)$, and the source proof’s strict-ball perturbation loses all uniform radius when $\|g\| = 1$.

A bounded novelty check searched the run indexes and arXiv for the source id, the later c_0 paper, “vector-valued spectrum”, “ c_0 ”, arbitrary or separable Banach target spaces, and large fibers. It found the exact $X = Y = c_0$ theorem in arXiv:1909.05105 and the distinct uniform-algebra Hilbert space paper arXiv:2102.06771, but no statement extending Theorem 2.4 to arbitrary Y or transferring it through a complemented c_0 . This is a bounded search, not a definitive novelty certification.

Human-review recommendation. Review as a likely valid partial result. The highest-value checks are that the scalar construction used in the two cases of [2, Theorem 2.4] has no hidden dependence on the target domain, and that its weak-star ultrafilter limit remains exactly the general $\mathcal{M}_\infty(B_{c_0}, B_Y)$ compactness described in [1].

Since this fiber is a singleton $\{\delta_{g(y)}\}$, we easily infer that $\mathcal{F}(g) = \{C_g\}$. If, whereas, $g \equiv x_0$ with $x_0 \in S_X$, the result follows from Proposition 5.2. $\square$

A lot of research is available in the literature about the basic simplest case $X = Y = \mathbb{C}$ (that is, homomorphisms from $\mathcal{H}^\infty(\mathbb{D})$ into $\mathcal{H}^\infty(\mathbb{D})$). Anyway, through our construction we can give a slightly different description of the spectrum $\mathcal{M}_\infty(\mathbb{D}, \mathbb{D})$ resembling the classical scalar-valued situation. Indeed, this vector-valued spectrum is projected onto $\overline{B_{\mathcal{H}^\infty(\mathbb{D})}}$, being one-to-one over the set $\{g \in \overline{B_{\mathcal{H}^\infty(\mathbb{D})}} : g(\mathbb{D}) \subset \mathbb{D}\}$. The remaining fibers (i. e. those over constant functions g of modulus 1) are large, they have plenty of non scalar-valued homomorphisms and each one contains an analytic copy of $B_{\mathcal{H}^\infty(\mathbb{D})}$.

For any infinite dimensional Banach space X we know from [2, Theorem 11.1] that each fiber of the spectrum $\mathcal{M}_\infty(B_X)$ contains a homeomorphic copy of $\beta(\mathbb{N})$. This canonically translates to fibers over constant functions $g \in \overline{B_{\mathcal{H}^\infty(B_Y, X^{**})}}$ in the spectrum $\mathcal{M}_\infty(B_X, B_Y)$. We can extend this result to fibers over (non-constant) functions g of constant norm 1. Recall that $\beta(\mathbb{N}) \setminus \mathbb{N}$ contains a homeomorphic copy of $\beta(\mathbb{N})$ so it is enough to obtain a homeomorphic copy of $\beta(\mathbb{N}) \setminus \mathbb{N}$ inside the fiber.

Proposition 5.4. *If X is an infinite dimensional Banach space and $g \in \overline{B_{\mathcal{H}^\infty(B_Y, X^{**})}}$ is a function of constant norm 1, then the fiber in $\mathcal{M}_\infty(B_X, B_Y)$ over g contains a homeomorphic copy of $\beta(\mathbb{N})$.*

Proof. Let $g \in \overline{B_{\mathcal{H}^\infty(B_Y, X^{**})}}$ be a function of constant norm 1 and fix $y_0 \in B_Y$. Since $\|g(y_0)\| = 1$, by [2, Theorem 10.5], for each $(r_n)_n \in \mathbb{D}$ with $r_n \rightarrow 1$ the sequence $(r_n g(y_0))$ has an interpolating subsequence for $\mathcal{H}^\infty(B_X)$ (which we still call $(r_n g(y_0))$). We now write $g_n = r_n g$ and consider the mapping

$$\begin{aligned} I : \mathbb{N} &\rightarrow \overline{\{C_{g_n}\}}^{w*} \subseteq \mathcal{M}_\infty(B_X, B_Y) \\ m &\mapsto C_{g_m}. \end{aligned}$$

By the universal property of $\beta\mathbb{N}$, there is a continuous extension $\beta I : \beta\mathbb{N} \rightarrow \overline{\{C_{g_n}\}}^{w*}$ such that $\beta I|_{\mathbb{N}} = I$. Since $(r_n g(y_0))$ is an interpolating sequence, the composition $\delta_{y_0} \circ \beta I$ is injective and so must be βI .

Finally, a straightforward computation shows that for every $\eta \in \beta(\mathbb{N}) \setminus \mathbb{N}$, the image $\beta I(\eta)$ lies in the fiber over g . $\square$

From the above, we know that, when X is infinite dimensional, fibers of $\mathcal{M}_\infty(B_X, B_Y)$ over constant functions or over functions of constant norm 1 are large. It is thus natural to ask whether the same is true for the remaining fibers, that is, fibers over non-constant functions $g \in \overline{B_{\mathcal{H}^\infty(B_Y, X^{**})}}$ with $g(B_Y) \subset B_{X^{**}}$. We have no general answer to this question. We can only say something under the hypothesis of existence of a polynomial that is not weakly continuous on bounded sets, and even in this case we can just reach the fibers over functions $g \in B_{\mathcal{H}^\infty(B_Y, X^{**})}$, as we present below. We do not know whether this result can be extended to fibers over functions g of norm 1 such that $g(B_Y) \subset B_{X^{**}}$ or not.

Supporting theorem evidence

Figures 2 and 3 reproduce PDF pages 9–10 of arXiv:1909.05105, containing Lemmas 2.2–2.3, Theorem 2.4, and the part of its proof generalized above.

References

- [1] V. Dimant and J. Singer, *A fibered description of the vector-valued spectrum*, arXiv:1902.01387 (2019); Math. Nachr. 293 (2020), 2104–2124.
- [2] V. Dimant and J. Singer, *Homomorphisms between algebras of holomorphic functions on the infinite polydisk*, arXiv:1909.05105 (2019); J. Math. Anal. Appl. 489 (2020), 124131.

The problem is that the construction of Φ_z^η in Theorem 1.1 is dependent on z (specifically, whether z has a subsequence whose coordinates are far away from 1 and which of those coordinates are used). Thus, to make formula (4) work we need all $z = g(x)$ to be of the *same* kind, independently of $x \in B_{c_0}$. In other words, we need (perhaps after a rotation) a subsequence $(g(x))_{n_k}$, independent of x with all of its elements at a positive distance from 1. Before proving the existence of such subsequence, let us first recall some information about the functions g in $\overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$.

Remark 2.1. Let $g \in \overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$. If there exists $x_0 \in B_{c_0}$ such that $\|g(x_0)\| = 1$ then $\|g(x)\| = 1$ for all $x \in B_{c_0}$. Hence, there are two alternatives for the range of g :

- $g(B_{c_0}) \subset B_{\ell_\infty}$.
- $g(B_{c_0}) \subset S_{\ell_\infty}$.

A function $g \in \overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$ can be viewed as a sequence of functions $g = (g_n)_n$ with each $g_n \in \overline{B}_{\mathcal{H}^\infty(B_{c_0})}$. Let us see now that the expected behaviour of g actually holds.

Lemma 2.2. *Let $g \in \overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$, $g = (g_n)_n$. Then at least one of the following occur:*

- (i) *There exists a subsequence (g_{n_k}) such that $g' = (g_{n_k})_k$ satisfies $g'(B_{c_0}) \subset B_{\ell_\infty}$.*
- (ii) *There exist $\lambda \in \mathbb{T}$ and a subsequence (g_{n_k}) such that $g_{n_k}(x) \rightarrow \lambda$ for all $x \in B_{c_0}$.*

Proof. Applying [11, Th. 2.17] for the sequence (g_n) we know that there exist a subsequence $(g_{n_k})_k$ and a function $h \in \overline{B}_{\mathcal{H}^\infty(B_{c_0})}$ such that $g_{n_k}(x) \rightarrow h(x)$, for all $x \in B_{c_0}$. If g does not satisfy (i), taking into account what we comment in Remark 2.1, we realize that

$$\sup_{k \geq k_0} |g_{n_k}(x)| = 1 \text{ for all } x \in B_{c_0}, \text{ and } k_0 \in \mathbb{N}.$$

This implies that there is $\lambda \in \mathbb{T}$ such that $h(x) = \lambda$, for all x and thus $(g_{n_k})_k$ satisfies (ii). $\square$

We also need to have an isometry between certain fibers of the vector-valued spectrum, in the spirit of what we stated in Remark 1.3 for the scalar-valued case.

Lemma 2.3. *Let $g, h \in \overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$ such that $|g_n(x)| = |h_n(x)|$, for all $n \in \mathbb{N}$ and $x \in B_{c_0}$. Then:*

- (i) *There exists a sequence (λ_n) in $\mathbb{T}$ such that $h_n(x) = \lambda_n g_n(x)$ for all $n \in \mathbb{N}$ and $x \in B_{c_0}$.*
- (ii) *The fibers $\mathcal{F}(g)$ and $\mathcal{F}(h)$ are Gleason isometric.*

Proof. (i) For a given n , if $g_n \equiv 0$ the result is trivial. If not, we can find an open subset V_n of B_{c_0} such that $g_n \neq 0$ on V_n . This implies that $\frac{h_n}{g_n}$ is a holomorphic function in V_n having constant modulus 1. This in turn implies that this function is constant and so (i) holds.

(ii) Given numbers (λ_n) in $\mathbb{T}$, let $\theta : B_{c_0} \rightarrow B_{c_0}$ be the mapping defined by $\theta(x_n) = (\lambda_n x_n)$. A similar argument as the one used in Remark 1.3, this time using [14, Prop. 5.3] instead of [3, Prop. 1.6] yields the desired result. $\square$

Now, we have all the necessary ingredients to proceed with the promised result.

Theorem 2.4. *For every $g \in \overline{B}_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$, there is a map*

$$\Psi_g : \beta(\mathbb{N}) \setminus \mathbb{N} \times B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} \rightarrow \mathcal{F}(g),$$

satisfying

- (1) *For all $\eta \in \beta(\mathbb{N}) \setminus \mathbb{N}$, the mapping $\Psi_g^\eta : B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} \rightarrow \mathcal{F}(g)$, given by $\Psi_g^\eta(h) = \Psi_g(\eta, h)$, is an analytic Gleason isometry.*

Figure 2: The coordinate dichotomy and the start of Theorem 2.4.

(2) If $\eta_1 \neq \eta_2$ are in $\beta(\mathbb{N}) \setminus \mathbb{N}$, then the images $\Psi_g^{\eta_1}(B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)})$ and $\Psi_g^{\eta_2}(B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)})$ lie in different Gleason parts.

Proof. Let $g \in \overline{B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}}$, $g = (g_n)_n$. We first assume that (g_n) has a subsequence (g_{n_k}) such that $g' = (g_{n_k})_k$ verifies $g'(B_{c_0}) \subset B_{\ell_\infty}$. We write $\mathbb{J} = \{n_k\}_k$, take the functions $\Phi_{z, \mathbb{J}}^k(w)$ from (3) and define

$$\begin{aligned} \Psi_g : \mathbb{N} \times B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} &\rightarrow \mathcal{M}_\infty(B_{c_0}, B_{c_0}) \\ \Psi_g(k, h)(f)(x) &= \Phi_{g(x), \mathbb{J}}^k(h(x))(f). \end{aligned}$$

For each $k \in \mathbb{N}$, the mapping $[(x, x') \mapsto \Phi_{g(x), \mathbb{J}}^k(h(x'))]$ is separately holomorphic in $B_{c_0} \times B_{c_0}$, and, by Hartogs' Theorem, it is holomorphic in $B_{c_0} \times B_{c_0}$. By restricting the previous mapping to the diagonal we obtain that $\Psi_g^k(h)(f) = \Psi_g(k, h)(f)$ is holomorphic. A straightforward computation shows that it is also bounded and that Ψ_g^k is well-defined. By the weak-star compactness of $\mathcal{M}_\infty(B_{c_0}, B_{c_0})$ (see for instance [14, p. 3]), for each fixed $h \in B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)}$, the mapping $\Psi_g(-, h) : \mathbb{N} \rightarrow \mathcal{M}_\infty(B_{c_0}, B_{c_0})$ has a unique extension to $\beta(\mathbb{N})$; this induces

$$\Psi_g : \beta(\mathbb{N}) \times B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} \rightarrow \mathcal{M}_\infty(B_{c_0}, B_{c_0}).$$

Note that for every $\eta \in \beta(\mathbb{N}) \setminus \mathbb{N}$, and $x \in B_{c_0}$ we have the following equality

$$(5) \quad \delta_x \circ \Psi_g^\eta(h) = \Phi_{g(x), \mathbb{J}}^\eta(h(x)).$$

Since we know from Theorem 1.1 that $\pi(\Phi_{g(x), \mathbb{J}}^\eta(h(x))) = g(x)$, the previous equality yields that $\xi(\Psi_g^\eta(h)) = g$ and thus the image of Ψ_g^η is contained in $\mathcal{F}(g)$. It is also readily seen that Ψ_g^η is a Gleason isometry, as we have that

$$\rho(\delta_x \circ \Psi_g^\eta(h), \delta_x \circ \Psi_g^\eta(h')) = \rho(\Phi_{g(x), \mathbb{J}}^\eta(h(x)), \Phi_{g(x), \mathbb{J}}^\eta(h'(x))) = \rho(\delta_{h(x)}, \delta_{h'(x)}),$$

where the last equality derives from the fact that, by Theorem 1.1, $\Phi_{g(x), \mathbb{J}}^\eta$ is a Gleason isometry. If we now take $\eta_1 \neq \eta_2 \in \beta(\mathbb{N}) \setminus \mathbb{N}$, it follows from (5) paired with Theorem 1.1 that the images $\Phi_{g(x), \mathbb{J}}^{\eta_1}(B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)})$ and $\Phi_{g(x), \mathbb{J}}^{\eta_2}(B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)})$ lie in different Gleason parts.

Additionally, for any fixed $f \in \mathcal{H}^\infty(B_{c_0})$, $x \in B_{c_0}$ and $\eta \in \beta(\mathbb{N}) \setminus \mathbb{N}$, the equality in (5) shows that the mapping $\delta_x \circ f \circ \Psi_g : B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} \rightarrow \mathbb{C}$ can be seen as the composition of the following analytic mappings

$$\begin{array}{ccc} B_{\mathcal{H}^\infty(B_{c_0}, \ell_\infty)} & \rightarrow & B_{\ell_\infty} \\ h & \mapsto & h(x) \end{array} \quad \begin{array}{ccc} B_{\ell_\infty} & \rightarrow & \mathbb{C} \\ z & \mapsto & \Phi_{g(x), \mathbb{J}}^\eta(z)(f), \end{array}$$

showing that Ψ_g^η is analytic.

If instead there is no subsequence such that the corresponding g' satisfies $g'(B_{c_0}) \subseteq B_{\ell_\infty}$, then by Lemma 2.2, we can find a subsequence (g_{n_k}) and a number $\lambda \in \mathbb{T}$ such that $g_{n_k}(x) \rightarrow \lambda$ for every $x \in B_{c_0}$. If $\lambda \neq 1$, for each $x \in B_{c_0}$ there exist $\delta = \delta(x) > 0$ and $k_0 = k_0(x) \in \mathbb{N}$ such that $|g_{n_k}(x) - 1| > \delta$ for all $k \geq k_0$. Let $\mathbb{J} = \{n_k : k \in \mathbb{N}\}$. Taking then

$$\Psi_g^k(h)(f)(x) = \Phi_{g(x), \mathbb{J}}^k(h(x))(f),$$

and proceeding as before we obtain the desired result. Finally, the remaining case, that is, $\lambda = 1$, follows from the previous argument combined with Lemma 2.3. $\square$

Remark 2.5. In [4, Lem. 2.9] it was shown that, for each $b \in \mathbb{D}$ and $w \in \overline{B_{\ell_\infty}}$, the fibers over w and (b, w) are homeomorphic. Then, it is observed in [3] that this homeomorphism is in fact, a Gleason isometry.

Figure 3: The conclusion and proof of Theorem 2.4.